INVIDIA CORE

Power protection that disappears into the wall

Track 1 · Reimagining the Stabilizer for the Next Decade · V-Guard Big Idea Tech Design Contest 2026

Every Indian home with a stabilizer has **two boxes doing one job in the same place** — a distribution board holding the MCBs and RCCB, and a stabilizer either plugged in behind the air conditioner or bolted to the wall beside the board.

V-Guard already manufactures both halves. **Invidia+** is their next-generation smart distribution board: it displays live voltage and RCCB status. **VMT 500/1000 Plus** are their mainline stabilizers. The company sells the box that *measures* voltage separately from the box that *corrects* it — to the same customer, for the same wall.

INVIDIA CORE collapses them into one: MCB, RCCB, surge protection and solid-state voltage regulation on a single DIN rail inside a single enclosure. Power protection stops being an appliance the customer buys and becomes infrastructure the building already has.

Why it hasn't been done — and how to make it work

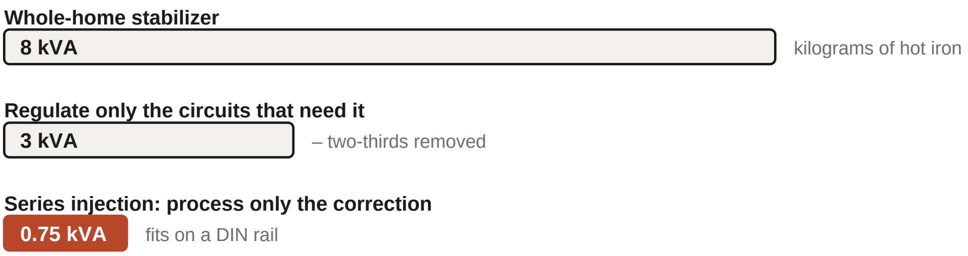
The objection is heat and weight. A whole-home relay-tap autotransformer for an 8–10 kVA connection is several kilograms of copper and iron dissipating real power inside a sealed plastic box. That cannot work, and we are not proposing it. Two reductions make the concept feasible.

Reduction 1 — regulate circuits, not homes.

The distribution board is **the only location in the home where selective per-circuit conditioning is physically possible.** A board already splits the supply into separate outgoing ways. So regulate the air conditioner and refrigerator circuits — real motors, real compressors — and leave lighting, fans and sockets alone, because LED drivers and BLDC fans are universal-input and do not need it. Roughly two-thirds of the requirement disappears before any design work begins.

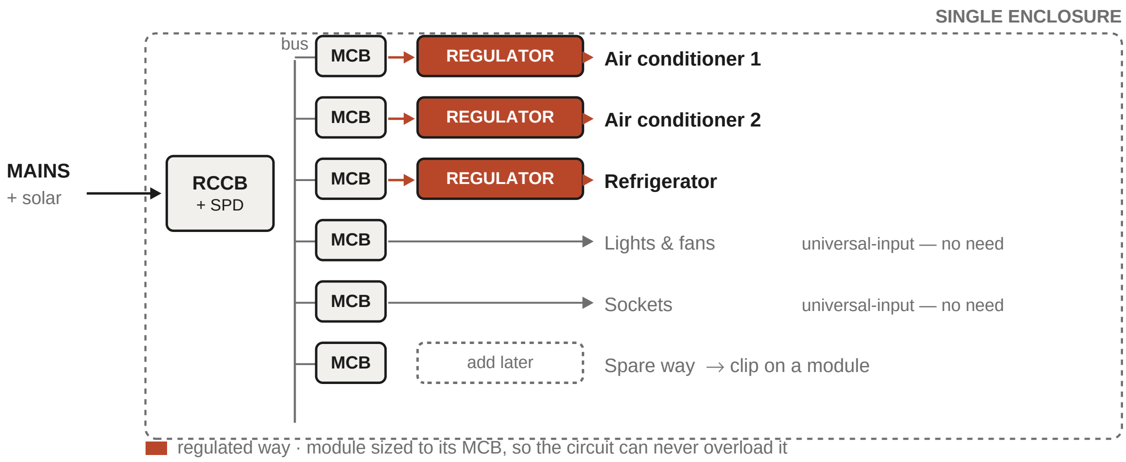
Reduction 2 — process the correction, not the load.

Rather than passing full load through a transformer, a **series voltage-injection compensator** is wired in series and synthesises only the correction voltage. Correcting ±25% on a 3 kVA circuit needs a converter of roughly 750 VA. This class of direct AC–AC compensator with step-up and step-down capability is well documented, and field evaluation of solid-state voltage regulators in low-voltage networks reports improved voltage profile with about 2.3% lower losses.



From roughly 8 kVA of hot iron to under 1 kVA of solid-state switching — the difference between impossible-in-a-board and fits-on-a-rail.

It also delivers what a tap-changer physically cannot: **continuous correction with no voltage steps, no dead-band and no relay clicking** — and the same silicon can perform surge clamping and power-factor correction.



Selective regulation: only the ways that need it carry a module. Spare ways accept modules later without rewiring.

Sizing and scalability

There is deliberately **no central stabilizer inside the board** — only N independent per-way modules. That resolves the three questions this architecture always attracts.

- **How is capacity guaranteed if a homeowner overloads a circuit?** Each module is rated to *the MCB protecting its way*, not to expected load. A 16 A MCB means that circuit physically cannot exceed 16 A. The protective device already in the enclosure *is* the capacity guarantee.
- **What if a regulated circuit also carries lights and minor loads?** Series injection processes only the correction fraction of whatever current actually flows, so a few hundred watts of incidental load costs almost nothing. Regulating it is unnecessary, not harmful.
- **What happens when the customer buys another air conditioner?** The electrician clips on another module — no board replacement, no rewiring of existing circuits. This requires reserved DIN width and a pre-wired regulator bus from day one. Because capacity is per-way rather than pooled, **there is no central rating to outgrow.**
- **Overload behaviour.** On overcurrent, over-temperature or excess inrush the module reverts to unregulated pass-through and logs the event. The circuit keeps working. The regulator must never be the component that trips the house.

Why now

Two new problems arrive at exactly this location. Rooftop solar under PM Surya Ghar has crossed **50 lakh installations**, against 7.94 lakh in the previous decade, creating reverse power flow and feeder-end voltage rise that a one-directional appliance stabilizer was never designed for. And home EV charging is straining household electrical envelopes — an AEEE/Kazam study warns residential charging could become the biggest hurdle to India's EV push, citing inadequate sanctioned loads, poor earthing and ageing wiring. **Both are board-level problems, not appliance-level ones.** And with air-conditioner penetration still around 8% of households, the installed base of vulnerable loads is only beginning to be built.

Novelty, stated honestly

We invented none of the constituent technologies, and the proposal is stronger for saying so. Solid-state stabilizers are commercial — Sollatek's AVR handles –30% to +22% input, correcting at 1250 V/s. DIN-rail stabilizers exist, but as generic relay-based industrial modules, not integrated residential consumer units, and not from any mainstream Indian brand. Series-injection compensation is well published. Smart Boards exist, including V-Guard's own Invidia+ — **but they measure and report; none correct.**

Our contribution is therefore four specific things: the **integration** (no one has combined protection and correction in one certified residential unit); the **per-circuit architecture** that makes it thermally and economically feasible; the **channel repositioning** below; and **portfolio fit** — V-Guard already makes stabilizers, switchgear, boards and cable. For almost any competitor this would be an acquisition. For V-Guard it is an integration.

Business value — the channel argument

A stabilizer is a **retail afterthought.** The customer buys an air conditioner, remembers they need a stabilizer, walks into a shop and compares prices across brands. Low margin, high competition, brand-fragile. A distribution board is **specified by the electrician or builder during construction**, before the customer has an opinion, and is never price-shopped by the homeowner. There is no comparison moment.

> **Integrating the stabilizer into the board moves power protection out of the price-comparison aisle and into the electrician's specification decision.**

That is a channel V-Guard already owns through switchgear and wires. It captures the customer years earlier — at wiring stage rather than appliance-purchase stage — and the attachment is durable, because nobody removes a distribution board to save Rs 1,500. This matters because V-Guard's problem is margin, not volume: FY25-26 revenue grew 7.0% to Rs 5,966 crore while PAT fell 1.7% to Rs 308 crore. The stabilizer market is growing — USD 770.65 million in 2025 to USD 1,202.60 million by 2034 — but on price.

For the consumer: one box instead of two, whole-circuit protection so a new appliance is covered automatically, no relay clicking, and nothing on the wall.

Risks and honest limitations

- **Thermal management is make-or-break.** If the enclosure cannot hold MCB ambient within its derating curve, the concept fails. The per-circuit and partial-power choices exist to make this tractable, and the thermal model in Phase 3 is built to answer it before any hardware is committed. Our working estimate is ~25–45 W dissipated per module, which puts the practical ceiling at **two to three regulated ways per enclosure** — not unlimited.
- **Certification.** Boards fall under IS 13032, stabilizers under IS 9815. A combined unit is effectively a new product category requiring BIS engagement — a real timeline cost, not a footnote.
- **No sub-circuit selectivity.** A board cannot see inside a circuit. If an air conditioner shares a way with lights, the remedy is a dedicated way — already standard practice for heavy loads — or a plug-in stabilizer for that one unit. Depth is the other physical constraint: the module must trade width for depth to fit a standard recess, making this a **wider board, not a deeper one**, with the front cover carrying the thermal path. Surface-mounted installations are unconstrained; concealed fitting needs a nine-inch wall.
- **Correction depth bounds the input range.** Converter size scales with how much voltage must be injected, so INVIDIA CORE targets roughly 170–290 V — not the 110–500 V of a mainline stabilizer. Homes with extreme excursions still need that separate box. Cost per kVA is also higher than relay-tap iron at low volume, so this launches premium.
- **The voltage problem is shrinking.** Grid quality is improving under RDSS and appliances tolerate wider ranges. The long-run case rests on the board becoming the home's power-quality and load-management node — which is why the architecture targets solar reverse flow and EV load management, not just low mains voltage.

Phase 3 plan — modelling first

The decisive question is thermal, and it is answerable analytically before hardware exists. We lead with simulation and reserve bench work for validation.

- **Lumped-parameter thermal model** of the enclosure: junction → heatsink → internal air → wall/front-cover paths, swept across module count (1–4), enclosure depth (55–95 mm) and vent area. Output: the maximum number of regulated ways that keeps MCB ambient inside its derating curve — the single most important number in the proposal.
- **Converter simulation** (SPICE / Python): series-injection waveforms for step-up and step-down operation, regulation versus input across 170–290 V, efficiency versus load, and step response contrasted against a relay tap-changer's dead-band.
- **Parametric study of the core trade-off:** converter VA versus correction depth, establishing where series injection stops being cheaper than a tap transformer — and therefore the product's honest input range.
- **Enclosure CAD plus an interactive configurator:** the wider, shallower geometry with a finned front-cover thermal path, confirming fit in a standard 75–95 mm recess, alongside a tool showing per-circuit module allocation, MCB-based sizing and cumulative heat budget.

– **Bench validation, if time permits:** a low-voltage scaled-down injection stage and thermocouple mapping in a real enclosure — the confirmation step, not the foundation of the argument.